Comparing the Effect of Ranibizumab vs. Aflibercept on Vision Acuity Outcomes for Neovascular AMD

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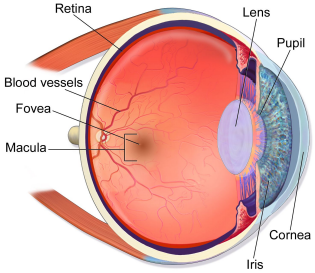
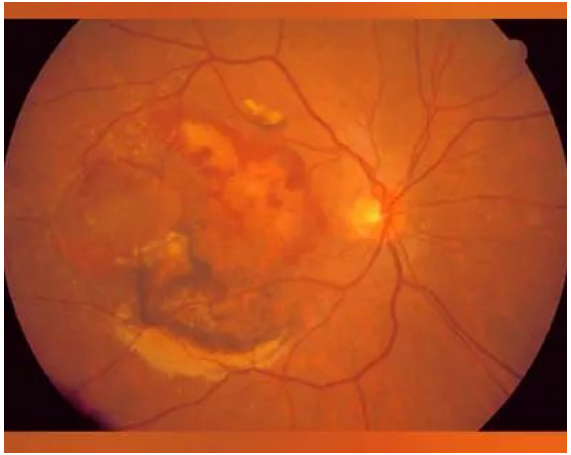
Neovascular Age-related Macular Degeneration

- One of the leading causes of irreversible central vision loss in older adults.
- Abnormal blood vessel formation in eye.
- Current treatments: Anti-VEGF injections to stop blood vessel formation.
- Approved treatments: Aflibercept (Eylea; Bayer) and Ranibizumab (Lucentis; Novartis), and some others.
- Extend MEH study by investigating causal effect of aflibercept vs ranibizumab on visual outcomes.



Eye Anatomy



Original Moorfields Study: What Was Studied

Study goal

- Real-world survival analysis of visual outcomes after anti-VEGF treatment for neovascular AMD
- Compared outcomes under **ranibizumab** and **aflibercept** at Moorfields Eye Hospital

Treatment setup

- Patients treated with either ranibizumab or aflibercept
- **Induction phase:** induction (or loading) phase comprised 3 injections at 1-month intervals. Patients must be inject 3 times in 90 days
- Patients who switched drugs were censored at the time of switch

Outcomes and analysis

- **Primary outcome:** time to VA ≥ 70
- **Secondary outcomes** included sustaining VA ≥ 70 and time to severe vision loss VA ≤ 35
- Analysis used Kaplan-Meier curves and Cox proportional hazards models

Insights from survival analyses during 12 years of anti-VEGF therapy for neovascular age-related macular degeneration

Fu, Dun Jack [1] ; Keane, Pearse [1]

Author affiliations

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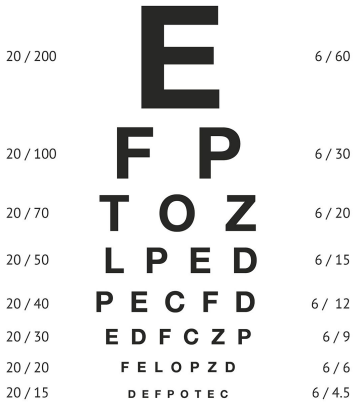
Data files

Nov 05, 2020 version files		20.33 MB
201025_anonymisation_SOP.docx		15.68 KB
201025_R_code_for_analysis.R		34.95 KB
MEH_AMD_survivaloutcomes_database.csv	Preview	20.27 MB
README_for_Moorfields_AMD_Database.rtf		2.33 KB

Click names to download individual files

Download full dataset



Original Moorfields Study: Cohort Construction

- Observational data queried for 10K eyes over 12 a year period (2008-2020).
- Inclusion criteria:
  - Only the first eye to receive treatment is sampled - in order to not violate IID assumption.
  - If both eyes were treated at the same time, then one is randomly sampled.
- Exclusion criteria:
  - Receiving photodynamic therapy, or anti-VEGF therapy before coming to MEH
  - Received other than aflibercept/ranibizumab (consistency assumption of SUTVA)
  - Unknown treatment date or VA value at baseline.

eTable. Demographics and Clinical Features of Cohort

	Overall	Aflibercept	Ranibizumab	Mean difference (95% CI)	P-value
Gender	n = 7802	n = 3951	n = 3851		
Female (%)	4776 (61.2)	2371 (60.0)	2405 (62.5)		.029
Male (%)	3026 (38.8)	1580 (40.0)	1446 (37.5)		
Age at baseline (Years)					
Mean (SD)	78.68 (8.75)	79.15 (8.70)	78.21 (8.77)	0.95 (0.55 - 1.33)	< .001
Ethnicity					
Afro Caribbean (%)	177 (2.3)	80 (2.0)	97 (2.5)		< .001
Caucasian (%)	4785 (61.3)	2081 (52.7)	2704 (70.2)		
Chinese (%)	41 (0.5)	17 (0.4)	24 (0.6)		
Mixed (%)	21 (0.3)	9 (0.2)	12 (0.3)		
South-East Asian (%)	652 (8.4)	350 (8.9)	302 (7.8)		
Unknown (%)	2126 (27.2)	1414 (35.8)	712 (18.5)		
VA at baseline in ETDRS letter score (approximate Snellen equivalent					
Mean	54.69 (20/80)	55.09 (20/80)	54.28 (20/80)	0.81 (0.10 - 1.52)	.025
SD	16.02 (3 lines)	15.76 (3 lines)	16.28 (3 lines)		
Treatment initiation					
Post-2013 (%)	4541 (58.2)	3951 (100.0)	590 (15.3)		< .001
Pre-2013 (%)	3261 (41.8)	0 (0.0)	3261 (84.7)		
Injection number					
Mean (SD)	10.25 (8.65)	12.11 (9.44)	8.34 (7.28)	3.77 (3.39 - 4.13)	< .001
Induction injection interval					
Mean (SD)	33.19 (34.14)	30.77 (6.16)	36.64 (52.94)	5.87 (3.74 - 8.0)	< .001
Induction phase					
Completed appropriately (%)	6212 (79.6)	3563 (90.2)	2649 (68.8)		< .001
Not completed (%)	1590 (20.4)	388 (9.8)	1202 (31.2)		

Original Moorfields Study: Results and limitations

Results

Primary outcome: How long until VA reaches ≥70 (20/40)?	56% reached VA ≥70; median time ≈ 2.0 years (95% CI 1.87 - 2.32)
Secondary outcome: How long is VA ≥70 sustained once achieved?	Among those reaching ≥70: median time sustained ≈ 1.1 years (95% CI 1.1 - 1.2)
Secondary outcome: How long until VA declines to ≤35 (20/200)?	Of the participants who started with a score >35, the median time to score under 35 was 8.7 years

Limitations

- Observational data → associations, not causation
- Informative Censoring: If a patient started on ranibizumab and then switched to aflibercept (or vice versa), all data after the switch was thrown out.
- Treatment era confounding. patients pre-2013 who only received ranibizumab and post-2013 received either ranibizumab or aflibercept.
- Hard to disentangle drug effect vs dosing protocol differences

Dataset Overview

MEH AMD Survival Outcomes Database · Moorfields Eye Hospital, London (2020)

Observational longitudinal cohort spanning 12 years at Moorfields Eye Hospital

One eye per patient (first treated eye; if simultaneous, randomly sampled)

Treatment era: Pre-2013 (ranibizumab only) and Post-2013 (either drug)

Patients who switched drugs censored at time of switch

Key Variables

Demographics: gender, ethnicity, age group at treatment initiation

Baseline VA: ETDRS letter score at first injection (mean = 54.7, SD = 16.0)

Induction phase: 3 injections within 90 days (loaded vs. not loaded)

Outcome VA: ETDRS letter score at each injection visit

Regimen: ranibizumab or aflibercept (no switching)

7,802
Total Patients

118,255
Injection Visit Records

3,951
Aflibercept Patients

3,851
Ranibizumab Patients

4,541
Post-2013 Patients

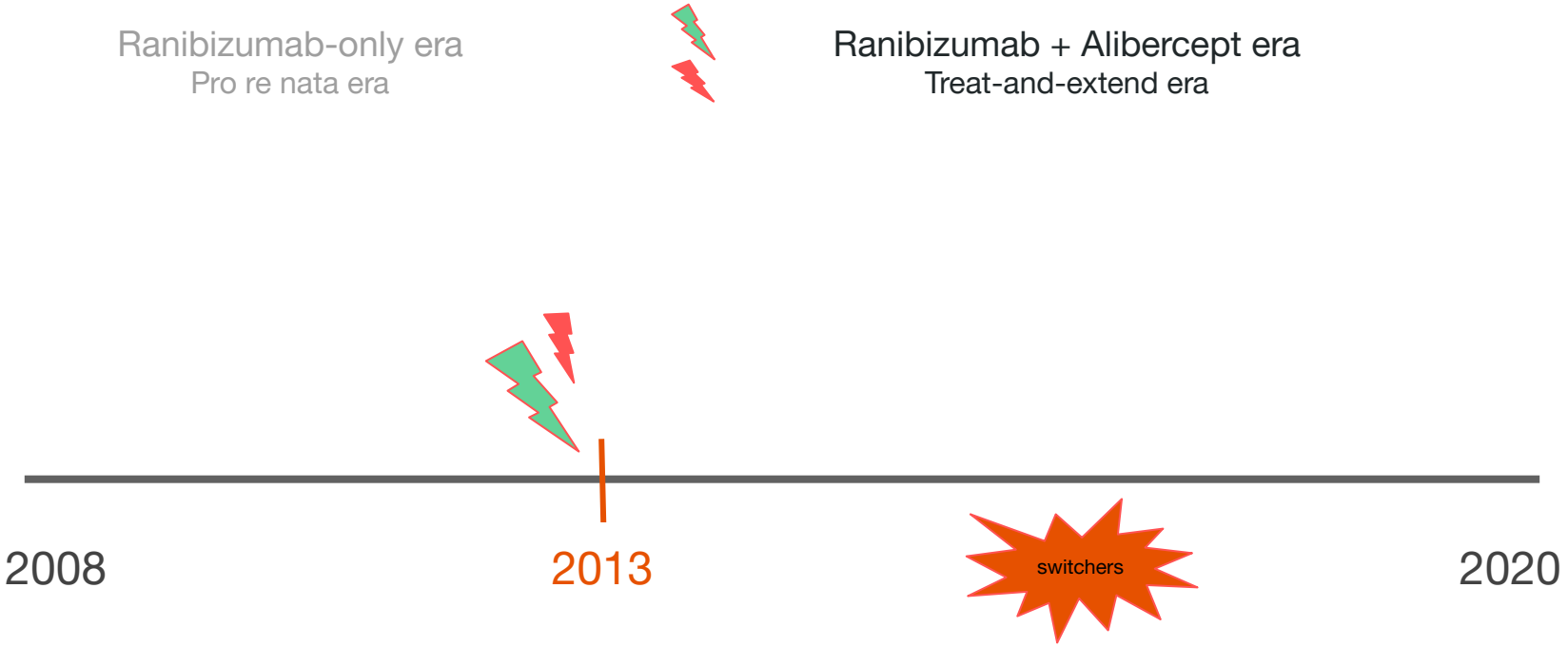
79.6%
Completed Induction

Causal Questions

1. **Population**—On average, does aflibercept cause faster visual recovery than ranibizumab—or does it just appear that way because of when and how each drug was used?

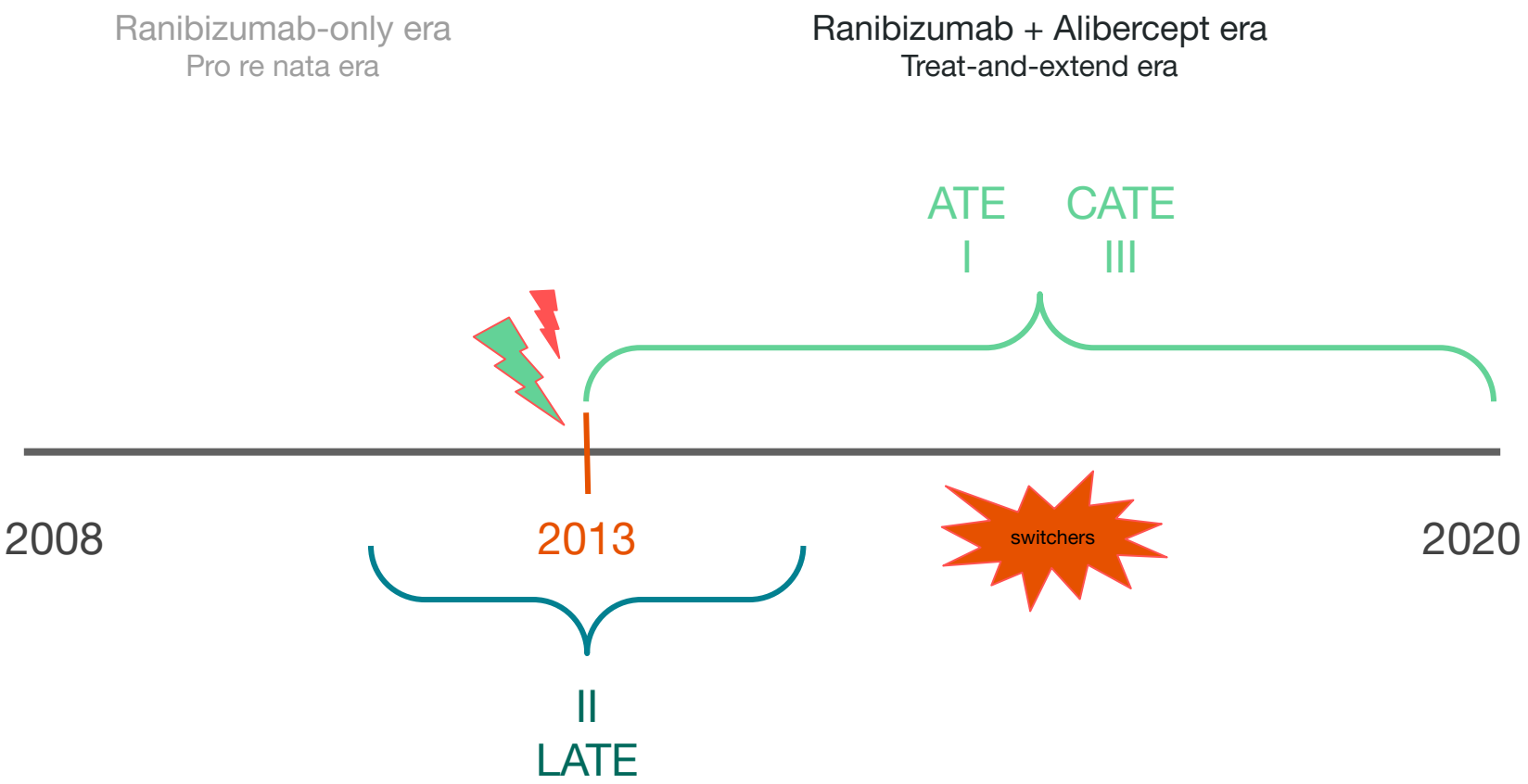
2. **Natural Experiment**—Among patients treated right around the moment aflibercept became available, did gaining access to it actually improve outcomes?

3. **Individual**—For a given patient (e.g. age, baseline vision, disease profile) which drug would produce the better outcome for them specifically?


Ranibizumab-only era
Pro re nata era
Ranibizumab + Alibercept era
Treat-and-extend era
2008
2013
switchers
2020

Methods

	Original source of failure	**Causal problem/opportunity**	**Methods**
I. Population	Treatment era confounding (pre/post-2013 protocol shift)	ATE estimation under selection bias	AIPW with survival outcomes
II. 'Natural experiment'	Aflibercept introduced at MEH in Oct 2013	Discontinuity in treatment access	Regression Discontinuity Design (fuzzy)
III. Individual	Population HR hides heterogeneous response	CATE across age × baseline VA space	TARNet / Dragonnet


Ranibizumab-only era
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ATE
I
CATE
III
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2020
II
LATE

Conclusion

What happens to patients on two anti-VEGF therapy? ⇒ What would happen if each treatment could be assigned optimally?

- leveraging a **causal estimate of aflibercept vs ranibizumab** ⇒ the number the original paper could not produce

- **identifying subgroups** of patients for whom both drugs are causally equivalent ⇒ eligible for the cheaper regimen option at no visual cost

- defining **patient-level treatment rule**: age + baseline VA + disease profile → drug recommendation at the point of prescription

References

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2. https://jamanetwork.com/journals/jamaophthalmology/fullarticle/2772986
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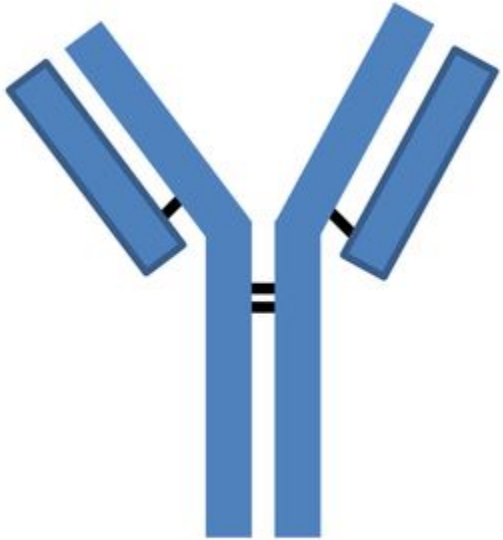
4. https://arxiv.org/abs/1906.02120

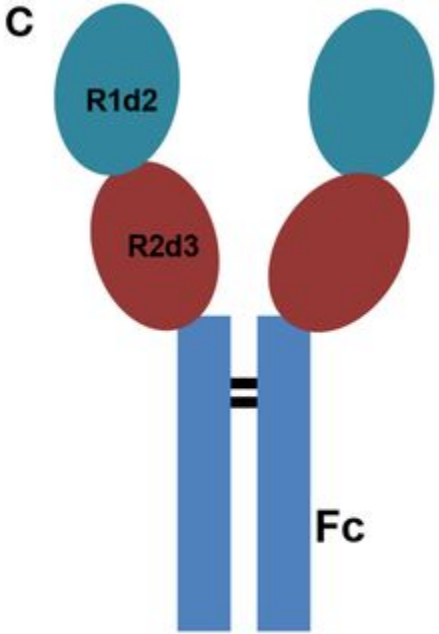
Supplemental

Annex

Ranibizumab
(Fab)



Bevacizumab
(Ab)



Aflibercept
(VEGF-trap)

Annex

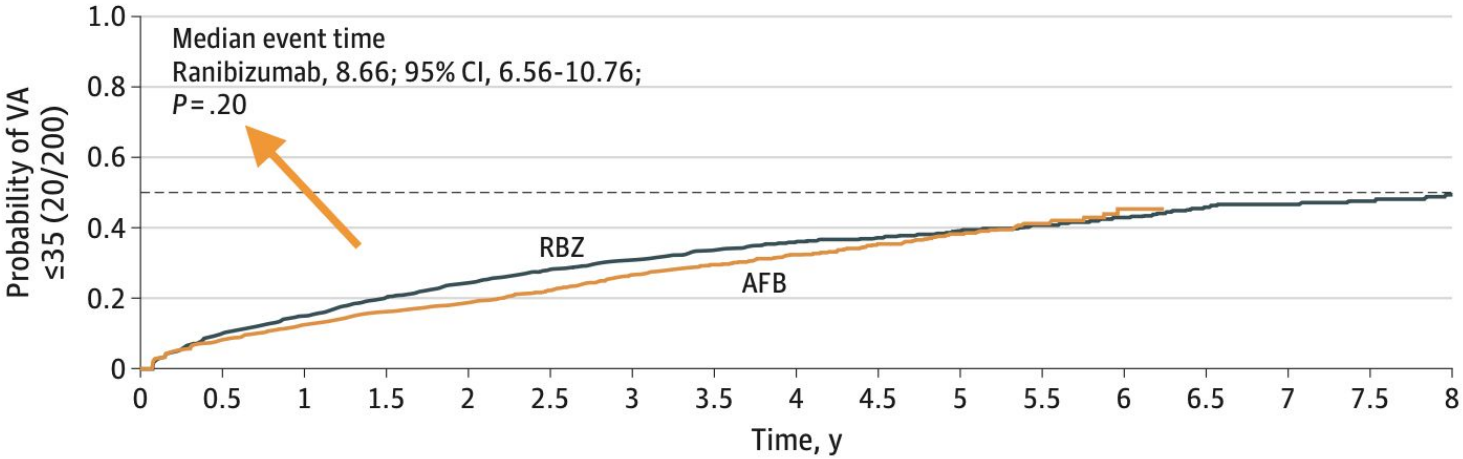
Table 2. Cox Proportional Hazards Regression Models for Visual Outcomes

Covariate	VA reaching ETDRS letter score ≥70 (20/40)		Failing to sustain VA ≥70 (20/40)		VA deteriorating ≤35 (20/400)	
	HR (95% CI)	*P* value	HR (95% CI)	*P* value	HR (95% CI)	*P* value
Sex						
Men	0.99 (0.91-1.07)	.75	1.01 (0.94-1.09)	.72	0.96 (0.86-1.07)	.42
Ethnicity						
White	0.82 (0.64-1.05)	.11	1.00 (0.79-1.26)	.99	1.31 (0.89-1.93)	.18
Chinese	0.97 (0.55-1.73)	.93	1.64 (0.94-2.87)	.08	1.33 (0.55-3.23)	.53
Mixed	1.21 (0.58-2.51)	.61	0.55 (0.26-1.14)	.11	1.71 (0.66-4.44)	.27
Southeast Asian	0.72 (0.55-0.94)	.02	1.10 (0.85-1.43)	.49	1.32 (0.87-2.01)	.19
Unknown	0.77 (0.60-0.99)	.04	1.18 (0.93-1.50)	.18	1.29 (0.86-1.91)	.21
Age at baseline	0.88 (0.86-0.90)	<.001	1.12 (1.10-1.15)	<.001	1.14 (1.11-1.18)	<.001
VA at baseline	1.43 (1.40-1.46)	<.001	0.89 (0.88-0.91)	<.001	0.71 (0.69-0.72)	<.001
Drug						
Ranibizumab	0.92 (0.77-1.11)	.39	1.07 (0.90-1.27)	.44	1.44 (1.13-1.84)	.003
Treatment start date before 2013	1.24 (1.03-1.50)	.20	0.96 (0.80-1.14)	.62	0.82 (0.64-1.06)	.13
Anti-VEGF injections	1.12 (1.10-1.15)	<.001	1.04 (1.02-1.05)	<.001	1.01 (1.00-1.02)	.17
Induction phase not completed	0.87 (0.77-0.98)	.02	1.04 (0.94-1.15)	.46	1.19 (1.03-1.38)	.02
Loading injection interval	1.00 (0.99-1.00)	.51	1.01 (1.00-1.01)	.97	1.00 (1.00-1.00)	.84
Concordance	0.763	NA	0.604	NA	0.738	NA
P value	NA	<.001	NA	<.001	NA	<.001

Abbreviations: ETDRS, Early Treatment Diabetic Retinopathy Study; HR, hazard ratio; NA, not applicable; VA, visual acuity; VEGF, vascular endothelial growth factor.

Annex

D Anti-VEGF agent



No. at risk

AFB	3352	2587	2097	1684	1360	1070	819	606	434	301	214	115	36	0	0	0	0
RBZ	3101	2216	1759	1376	1101	861	680	527	402	297	241	209	178	148	127	98	76